Quantum Reservoir Computing for Icequake Prediction Modeling

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Abstract

Basal slip is a cryoseismic event that occurs at the interface between a glacier and the underlying bed, often used as an analog for tectonic earthquakes. They are direct indicators of ice stream dynamics and mass loss, connecting to sea level rise projections on decadal timescales. We use the Whillans Ice Plain (WIP) in West Antarctica, a tidally paced stick-slip system with a physically documented bimodal regime structure and an 11-year Global Navigation Satellite System (GNSS) catalog of over 5,000 events, as a demanding benchmark for quantum-classical hybrid learning. Predicting the timing of Whillans stick-slip events remains an open challenge, complicated by small observational catalogs, non-stationary tidal forcing, and multimodal inter-event distributions.

We first establish a classical baseline by benchmarking XGBoost, CatBoost, and ridge regression for time-to-next-slip regression using tidal-phase features, previous event timing, and slip size. However, classical methods systematically fail on long-interval events and cannot resolve the fine tidal-phase structure needed to constrain rupture nucleation timing, inter-event stress transfer, and proximity to ice stream shutdown, motivating richer nonlinear feature representations.

We address this with a two-stage quantum-classical pipeline: a classical classifier separates 12- and 24-hour regimes before routing each to a dedicated fully connected transverse-field Ising (FC-TFI) quantum reservoir, whose Pauli expectation values encode cross-feature tidal correlations in a single circuit execution. Noiseless simulation achieves parity with the classical baseline, consistent with theoretical expectations. Recent IBM hardware results suggest that a measurable advantage should emerge under realistic noise, which we target in ongoing hardware evaluation.

1 Introduction

1.1 Whillans Ice Plain

The West Antarctic Ice Sheet contains enough ice to raise global mean sea level by approximately 3.3 meters if fully discharged, with ice streams serving as the primary means through which mass is transferred to the ocean [1]. Marine-terminating ice streams account for up to 90% of Antarctic ice sheet discharge [2], making their dynamics a first-order control on sea level projections over decadal and centennial timescales. Whillans Ice Plain (WIP), the downstream portion of Whillans and Mercer ice streams flowing into the Ross Ice Shelf, is unique among Antarctic ice streams in that its flow is dominated by discrete macroscopic stick-slip events; sudden (30-60 minute) displacements of approximately 0.4 meters occurring once or twice per day with little motion occuring between events [3], [4]. Each 30 min slip event releases energy equivalent to a moment magnitude 7 earthquake and is detectable globally through teleseismicty. [5].

While the physical mechanism is not well-constrained, there are known forcings: Slip events tend to occur during falling ocean tide, just after high tide, and just before low tide, with timing governed by the evolving strength of the ice-bed interface as described by rate-and-state friction [5], [6]. As well as, the stick-slip cycle is driven by the interplay between tidal forcing and basal friction, modulated by subglacial hydrology [3], [7], [8].

The tidal forcing produces a physically documented bimodal structure in inter-event intervals. The timing of slip is governed by the superposition of tidal constituents that modulate basal shear stress and determine whether the frictional threshold for slip is reached in a given tidal cycle [9]. During spring tides, two events occur per day at regular phase; during neap tides (tides with smaller tidal ranges), low-tide events are skipped, producing long inter-event intervals exceeding 20 hours, with this longer time between slips [10]. This skipped-slip mechanism is linked to the long-term deceleration of WIP at approximately 0.6% per year squared [11], which reduces the stressing rate and pushes the system closer to conditions where slip fails to nucleate. Numerical modeling suggests this deceleration will not reverse without a substantial increase in basal water supply, and that WIP may eventually stagnate analogously to the nearby Kamb Ice Stream, which shut down approximately 150 years ago [11], [12].

Katz et al. [4] assembled an 11-year (2008–2019) catalog of over 5,000 stick-slip events using GNSS data from 48 stations, demonstrating that slip-event frequency varies at fortnightly, monthly, and semiannual ocean tidal periods and that the secular decrease in event frequency reflects increased prevalence of one-slip diurnal cycles over two-slip cycles. This catalog is the dataset used in this study and represents one of the longest records of ice-stream seismicity.

The scientific questions this system raises exist at two timescales that motivate different prediction accuracy thresholds. Bulk questions, such as whether a given tidal cycle will produce one slip or two, and what the ice stream's total mass flux is over fortnightly or annual periods, could be evaluated from coarse tidal-phase structure and are scientifically useful for constraining ice dynamics models [9], [11]. Finer-scale questions demand tighter constraints on slip timing such as are consecutive slip events mechanically coupled through stress transfer in the subglacial till [7], [13], and what does the evolving distribution of inter-event intervals reveal about proximity to shutdown [9], [11]? We show that classical methods can provide prediction accuracies that could aid in answering the first class of questions, but systematically fail on the second. Thus, we develop a quantum-classical pipeline designed to improve prediction accuracy in an effort to better inform the second class of questions.

1.2 Classical Machine Learning

Machine learning has been applied to seismic event prediction across a range of problem types. In tectonic seismology, deep learning approaches, including neural temporal point processes and graph neural networks, have shown promise for aftershock forecasting and magnitude prediction from event catalogs [14], [15]. In cryoseismology specifically, machine learning has been applied to event detection and classification on WIP seismic records [16], but precise timing regression, specifically predicting the exact inter-event interval rather than detecting that an event occurred, remains unaddressed.

For tabular regression on small structured datasets, gradient-boosted tree methods represent the state of the art, consistently matching or outperforming deep learning approaches across benchmark suites [17]. XGBoost in particular has been applied to GNSS-derived geophysical time series, outperforming traditional interpolation methods by substantial margins when spatiotemporal feature correlations are incorporated [18]. Its properties, such as robustness to missing data, interpretable feature importance, and nonlinear response modeling, are well-suited to event-catalog regression where physical forcing variables are known, but their interactions are complex [19]. For WIP, the dominant forcing is ocean tides, and the primary features are tidal phase, previous inter-event interval, and slip magnitude [4], [6]. On the short-interval regime, where tidal phase and slip timing are tightly coupled [3], a well-tuned XGBoost model is expected to perform well.

The limitation arises in the bimodal structure of the inter-event distribution. The fundamental architecture of gradient-boosted trees, sequential binary splits over individual features, requires exponentially many splits to resolve higher-order nonlinear interactions between correlated predictors, and the resulting models are known to struggle with multimodal target distributions where training examples are asymmetrically distributed across modes [17], [20]. The long-interval regime is, by definition, the sparse tail of the training distribution [4], [9], and the stress-transfer conditions that determine whether a low-tide event will be skipped depend on a nonlinear interaction between tidal amplitude, residual stress, and mechanical bed state [7], [9] that trees cannot efficiently represent.

We benchmark XGBoost, CatBoost, and ridge regression using a shared upstream regime classifier that separates short- and long-interval events prior to regression, motivated by the bimodal structure documented by [4], [9]. This ensures that the classical and quantum pipelines below are evaluated on identical problem formulations. The performance gap, of more accurate prediction on short intervals than on long intervals, is consistent across all three classical methods, confirming it is structural rather than incidental to any single algorithm.

1.3 Quantum Reservoir Computing

Quantum kernel methods provide a theoretical framework for accessing feature representations that are exponentially richer than those available to classical methods. By encoding input data as quantum states and measuring Pauli expectation values, one implicitly defines a kernel that captures inner products in a 2^n-dimensional Hilbert space, meaning that a 6-qubit system implicitly computes all degree < 6 cross-features simultaneously through entanglement [21], [22]. The specific cross-feature interactions relevant to WIP slip timing, such as phase differences between tidal constituents modulating basal stress, appear naturally in ZZ correlator measurements after Hamiltonian evolution, without requiring explicit polynomial feature construction. This motivates quantum approaches for this problem, but the choice of architecture matters significantly. Variational quantum circuits (VQCs), the most commonly explored quantum ML architecture, train parameterized circuits by gradient descent. When gradient-based methods are used, gradient variance decays exponentially with qubit count or circuit depth, leading to the barren plateau phenomenon [23], widely considered one of the main limitations of variational quantum algorithms [24].

Quantum Reservoir Computing (QRC) avoids this by design [25]. The reservoir is fixed and untrained; only a classical linear readout is optimized, so training reduces to classical regression on the Pauli expectation values extracted from reservoir measurements, avoiding the need to compute gradients through quantum circuits and thereby avoiding barren plateaus. The FC-TFI reservoir architecture employed here is theoretically preferred for this setting: disordered inter-spin couplings maximize memory capacity by breaking permutation symmetry [26], while Hamiltonian symmetries suppress the exponential concentration of Pauli observables that renders random quantum circuit reservoirs uninformative [27]. Its all-to-all connectivity encodes cross- constituent tidal interactions in a single circuit execution that XGBoost would require exponentially many tree splits to approximate [17]. Hardware noise, counterintuitively, is expected to help rather than hurt. Under certain noise regimes, amplitude damping acts as a structured regularizer on quantum feature distributions, yielding better performance than noiseless reservoirs [28]. This is confirmed experimentally: on IBM Heron R2, a gate-based QRC implementation outperformed noiseless simulation on ENSO forecasting, with device noise concentrating variance in feature directions relevant to the readout [29], [30] demonstrated 12–34% improvement over classical baselines on tabular financial prediction on real IBM Heron hardware. An advantage that is notably absent in noiseless simulation and not seen in simulations with classical noise injection. Therefore, the target is hardware execution under realistic noise, where structured decoherence is expected to produce the advantage that simulation obscures.

1.4 Contributions

This paper makes three contributions, each addressing a distinct gap. First, it provides the first application in the literature of machine learning, classical or quantum, to WIP slip-timing regression on the [4]. Prior cryoseismological ML work has focused on event detection and classification within shorter observational windows [16], [31], rather than timing regression on a decadal record with documented non-stationarity. Second, it introduces a physically motivated quantum-classical pipeline whose architecture is grounded in the documented physical structure of the WIP system. The two-stage design with regime classification followed by regime-specific FC-TFI quantum reservoirs directly reflects the bimodal inter-event distribution [4], [9]. The FC-TFI Hamiltonian is selected for its memory capacity and Pauli concentration properties [27]; the regime-specific readout is selected because pooling across both interval populations would suppress the performance gap that the architecture is designed to address. Third, it contributes an empirical data point to the open question of where QRC hardware advantage materializes on real scientific data. Noiseless simulation achieves parity with the classical baseline, consistent with theoretical expectations that noiseless quantum kernels span comparable function classes to classical methods [32]. Ongoing hardware evaluation targets the noise regime where structured decoherence is expected to produce an advantage [28], [30]. The result will highlight whether near-term quantum hardware can be applied to geophysical data to better constrain prediction timing.

2 Methods

2.1 Data

The dataset is derived from [4], which provides a GNSS-derived catalogue of roughly 5,150 stick-slip events on Whillans Ice Plain spanning 2008–2019. Six features were engineered for modeling [33], each physically motivated: four capture tidal state, tide machine-learning regression of tide event, and two capture slip history. This choice reflects the finding in [4] that slip timing is modulated by tidal forcing across multiple periodicities, making tidal state a primary physical driver rather than an incidental covariate. All models operate on a windowed representation of consecutive events, where the window length was selected empirically to minimize validation error.

2.2 Preprocessing and Baselines

A unified preprocessing pipeline was applied identically across all model types, including removal of events with ambiguous timing labels and standardization fit exclusively on training data to prevent leakage. The data were partitioned into training, validation, and test sets using stratified sampling. Two baselines were implemented to assess whether machine learning models were obtaining useful information from the features and had predictive capabilities beyond a simple auto correlation. The first baseline is a median which predicts the training-set median for every event regardless of input. The second is a persistent baseline which carries forward the most recent inter-event interval. The primary evaluation metric throughout is mean absolute error in seconds, chosen for robustness to the outliers.

2.3 Classical ML Models

The distribution of inter-event times is strongly bimodal, clustering around 12-hour and 24-hour periods that reflect the known coexistence of once-daily and twice-daily slip cycles depending on tidal forcing [4]. A model asked to cover both regimes simultaneously tends toward systematically wrong intermediate predictions. An XGBoost classifier therefore first routes each sample into a short or long regime, after which predictions are constrained to physically plausible bounds within each regime. This decomposition also simplifies the task presented to the quantum reservoir: rather than one reservoir spanning the full bimodal distribution, two independent reservoirs each specialize on a more coherent target range.

Three classical regressors, Ridge Regression, XGBoost, and CatBoost, were trained as comparators. Ridge provides a regularized linear baseline [34]. XGBoost and CatBoost both build sequential ensembles of decision trees, differing primarily in how they handle variance across the training sequence; CatBoost's approach makes it somewhat better suited to continuously varying nonlinear relationships, while XGBoost's conservative depth constraints reduce overfitting in the data-sparse long-interval regime [19], [35]. Hyperparameters for both boosting models were tuned against validation error using Optuna.

2.4 Quantum Reservoir

Following regime classification, two independent FC-TFI quantum reservoirs are instantiated, one per regime, each with distinct randomly sampled Hamiltonian parameters, ensuring they develop different quantum dynamics rather than redundant representations [25], [26]. The Transverse-Field Ising Hamiltonian is a suitable choice for this role because its ZZ coupling terms generate the entanglement structure needed to capture correlations across input features, while the transverse field drives quantum fluctuations that prevent the dynamics from collapsing into trivial classical behavior. Time evolution is approximated through Trotterization, trading exact evolution for a decomposition that remains implementable on near-term hardware without sacrificing the nonlinear dynamics. Expressive capacity is further extended through data re-uploading where the classical input vector is periodically reintroduced between layers of reservoir evolution. Because a single unitary encoding pass limits the model to a narrow frequency spectrum, re-uploading allows the circuit to approximate higher-order nonlinear functions [36]. At readout, the reservoir state is projected onto a set of local and pairwise observables rather to capture both single-qubit dynamics and multi-qubit correlations while keeping the feature dimensionality tractable. Temporal history across the event window is then embedded classically through an exponential decay weighting, with the decay rate tuned to each regime: faster decay for the short 12-hour cycle, where recent events dominate, and slower decay for the 24-hour cycle, where a longer historical context is physically meaningful. Finally, because the reservoir's random initialization introduces stochasticity into the feature space, the pipeline is instantiated multiple times and predictions are aggregated across the best-performing configurations, reducing sensitivity to any particular draw of Hamiltonian parameters.

2.5 Noisy Simulation

For the noisy simulation, we leveraged the `qiskit-aer` simulation with the `FakeFez` noise backend to accurately represent intrinsic noise on the `ibm_fez` node. Since we only needed to change the backend which is agnostic of the rest of the code, the experiment remained exactly the same as the noiseless simulation. Running a noisy simulation allowed us to pre-optimize the hyper parameters of the circuits in preparation for running on the available hardware.

3 Results

3.1 Window Size Selection

Prior to model evaluation, the effect of lookback window length on prediction performance was assessed by training the full classical pipeline across lookback sizes ranging from 10 to 50 prior events and evaluating validation MAE. Results are summarized in Table 1. Performance improved consistently from 50 prior events (MAE: 13,790 s) down to 20 prior events (MAE: 11,353 s), with larger windows likely introducing noise from temporally distant events whose relevance to current slip timing is negatively impacted by the long-term deceleration trend in the catalog. Performance degraded for lookback sizes smaller than 20, suggesting that at least 20 prior events are needed to capture the fortnightly tidal modulation structure. A lookback of 20 prior events was therefore selected for all models; each sample consists of the current event plus the 20 preceding events, yielding a 126-dimensional feature vector (21 total events $\times$ 6 features per event).

3.2 Regime Classifier Performance

The XGBoost regime classifier achieved an overall test accuracy of 83.9%, with 91 misrouted samples out of 565 total. Regime-level performance is reported in Table **??**. The short regime (true inter-event interval $<$65,000 s, $n = 367$) achieved precision, recall, and F1-score of 0.88. The long regime ($\geq$65,000 s, $n = 198$) achieved precision and recall of 0.77. The macro-averaged F1-score was 0.82.

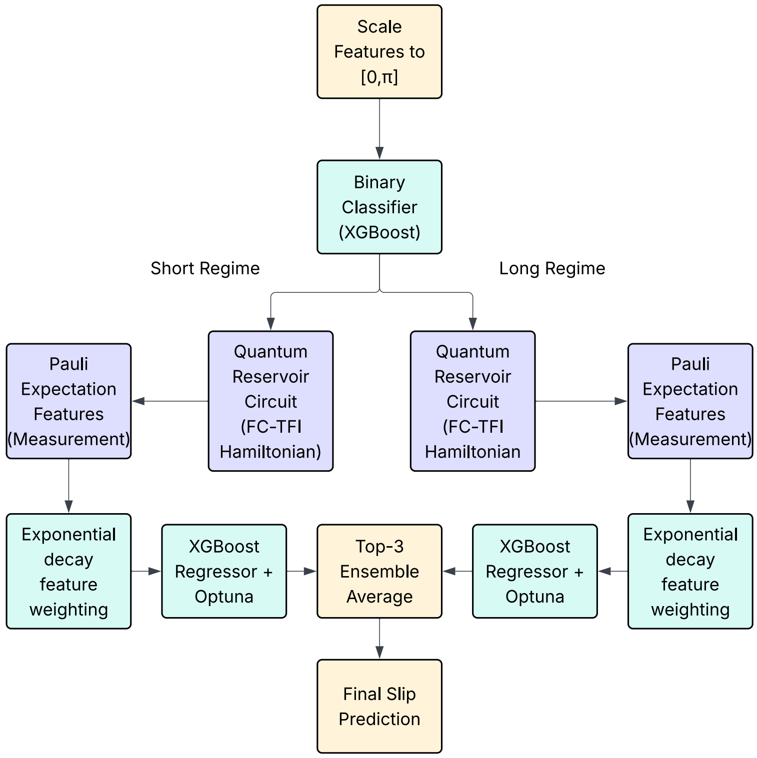

Figure 1: Complete quantum reservoir pipeline.

3.3 Classical Model Performance

Two classifier-naive baselines were computed on the test set. The median baseline achieved an MAE of 17,168 s and RMSE of 20,686 s. The persistence baseline achieved an MAE of 22,803 s and RMSE of 28,630 s.

All three classical regressors substantially outperformed both baselines. Full results are reported in Table 2. XGBoost achieved the best overall performance with a test MAE of 8,932 s and RMSE of 16,888 s, representing a 47.9% reduction in MAE relative to the median baseline. CatBoost and Random Forest performed comparably at MAE values of 9,211 s and 9,042 s respectively. Ridge regression was the weakest classical model at 9,702 s.

The regime-split breakdown for XGBoost is reported in Table **??**. On short-interval events ($n = 367$), XGBoost achieved an MAE of 7,769 s and RMSE of 15,324 s. On long-interval events ($n = 198$), performance degraded to an MAE of 11,089 s and RMSE of 19,457 s, a 42.7% increase in MAE relative to the short regime.

3.4 Noiseless QRC Performance

The noiseless FC-TFI quantum reservoir ensemble achieved a test MAE of 8,999 s, RMSE of 16,847 s, and R^2 of 0.30. Individual iterations varied in test MAE from 9,059 s to 9,215 s. The top-3 iterations selected by validation MAE (iterations 1, 5, and 2) were averaged to produce the ensemble result. Within the ensemble, short-regime validation MAE ranged from 3,808 s to 4,051 s across iterations, and long-regime validation MAE ranged from 4,014 s to 4,128 s.

The scatter plot in Figure 3 shows the distribution of true versus predicted time to next slip on the test set. The short-interval cluster shows dense alignment

Lookback (prior events)	**Validation MAE (s)**
10	12,188
15	12,192
19	11,583
20	**11,353**
21	11,777
30	11,579
40	12,762
50	13,790

Table 1: Validation MAE as a function of lookback window length (prior events only, excluding the current event). A lookback of 20 prior events was selected for all models.

Model	**MAE (s)**	**RMSE (s)**
Median baseline	17,168	20,686
Persistence baseline	22,803	28,630
Ridge Regression	9,702	17,275
CatBoost	9,211	16,985
Random Forest	9,042	16,986
XGBoost	**8,932**	**16,888**
QRC (noiseless)	8,999	16,847
QRC (noisy sim)	9,561	17,516
QRC (hardware)	—	—

Table 2: Test set performance for all models. Baselines are classifier-naive. Noisy simulation and hardware results are pending. Best classical result in bold.

near the perfect-prediction diagonal with low absolute error. The long-interval cluster is more dispersed, with a visible group of correctly routed long-interval events and a separate cluster of high-error points consistent with the 16.1% long-regime misclassification rate.

3.5 Noisy QRC Simulation

The MAE error recorded, ensembling the top-3 best performing reservoirs, was 9561 s, RMSE of 17,516 s, and R^2 of 0.24. Individual iterations varied in test MAE from 9621 s to 9764 s. The top-3 iterations selected by validation MAE (iterations 2, 3, and 4) were averaged to produce ensemble results. Within the ensemble, short-regime validation MAE ranged from 6768 s to 7050 s across iterations, and long-regime validation MAE ranged from 13,842 s to 14,077 s.

3.6 Hardware Run

The proposed quantum reservoir computing architecture was deployed on physical quantum hardware utilizing the IBM Quantum `ibm_fez` compute node. To extract the used expectation values, each quantum circuit was executed using the Qiskit Runtime Estimator primitive configured with 300 and 5 shots per circuit for our two attempted runs. Measurement error mitigation was enabled via Resilience Level 1, utilizing Twirled Readout Error Extinction [37] to suppress hardware-specific readout assignment errors.

Physical execution on noisy intermediate-scale quantum (NISQ) devices introduces strict resource limitations. In this study, hardware access was constrained to the 10-minute maximum execution window provided under the standard free-tier compute allocation. Due to these temporal limitations, compounded by the sequential processing of temporal events and the circuit depth inherent to the Trotterized reservoir dynamics, the physical deployment was severely bottle-necked. The experiment exhausted its compute allocation after successfully completing only two of the planned 21 ensemble iterations.

While the extraction of multi-qubit Pauli expectation values from the `ibm_fez` backend successfully generated the target hybrid feature space, the restricted iteration count severely limits the statistical robustness of the final model. In the idealized simulation, averaging across a 21-iteration ensemble of random reservoir initializations smooths variance, mitigates the risk of capturing a suboptimal projection space, and corrects for small hardware fluctuations. With only two iterations, the physical hardware performance metrics are heavily biased by the specific random seeds and the unmitigated gate infidelities present during that narrow execution window.

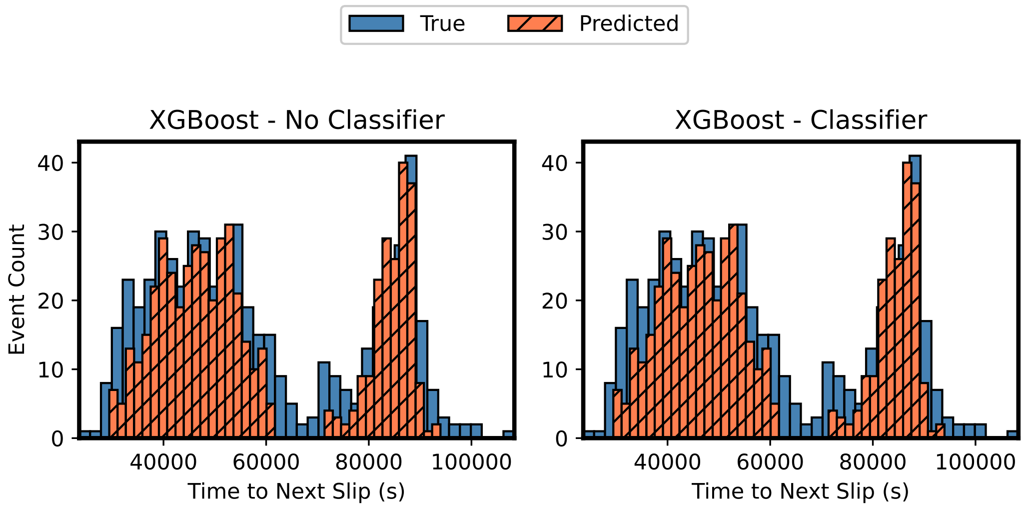

Figure 2: Histogram showing XGBoost performance before and after adding the classifier and time-constrained prediction.

Consequently, the preliminary hardware results represent a strict lower-bound estimation of the architecture's predictive capability. The successful execution of this limited subset proves the viability and feasibility of our quantum-classical framework on real superconducting hardware. However, scaling this methodology to achieve statistical convergence and rigorously benchmark it against classical models requires extended QPU access. Securing expanded computational resources is a necessary next step to fully realize and validate the advantages, or lack thereof, of this temporal quantum reservoir framework.

4 Discussion

4.1 Classifier Performance and Its Effect on Downstream Results

The regime classifier achieved a test accuracy of 83.9%, with meaningfully asymmetric performance across regimes: 88% F1 on short-interval events versus 77% F1 on long-interval events. This asymmetry is physically interpretable. Short-interval spring-tide events occur under high-amplitude, regular tidal forcing where the relationship between tidal state features and slip occurrence is stronger and more consistent [3], [9]. Long-interval neap-tide events, by contrast, arise when the tidal stress pulse falls near the frictional threshold, making the slip decision sensitive to residual stress state and subglacial mechanical conditions that are not directly observable in the feature set [7], [13]. The 16.1% long-regime misclassification rate therefore reflects a genuine physical ambiguity in the data rather than a tuning failure.

However, classifier errors set a structural floor on downstream regression performance. Misrouted long-interval samples are constrained to the short-regime prediction interval (25,000–65,000 s) and thus have large absolute errors. This effect is visible in Figure 3 as the high-error cluster of long-interval events with predicted values near the short-regime ceiling. Improving long-regime classification precision through additional features (such as ice sheet velocity or thickness), a longer or more continuous data set, or

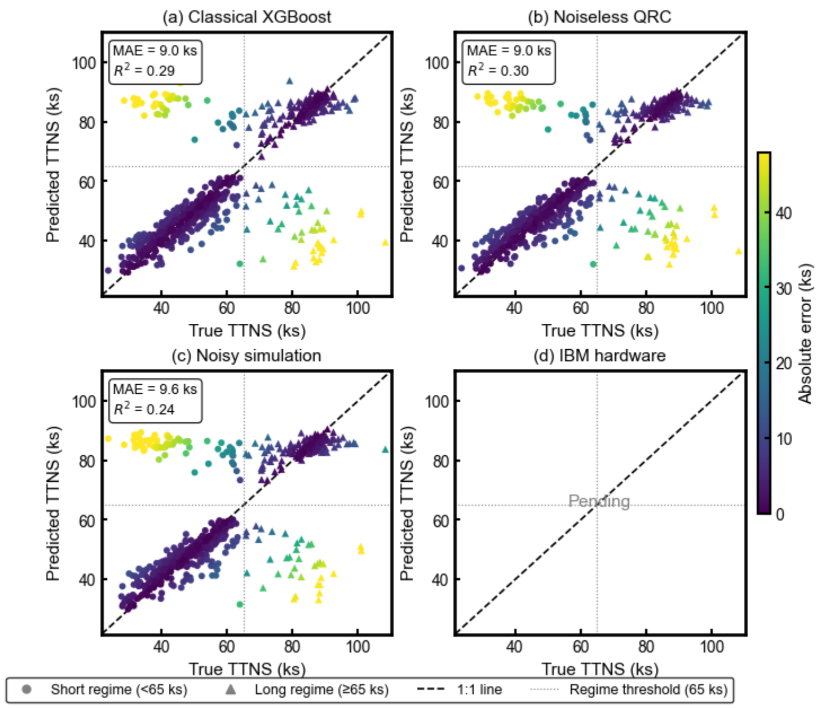

Figure 3: True versus predicted plots for (a) Classical XGBoost, (b) the noiseless quantum reservoir architecture, and (c) the same quantum reservoir architecture as (b) with the IBM hardware noise backend. Once hardware results are obtained, they will be shown in (d).

incorporation of subglacial hydrological proxies would be a good avenue in future work for improving overall pipeline performance independently of the quantum reservoir architecture.

4.2 Classical ML Model Performance

All classical regressors substantially outperformed the classifier-naive baselines, confirming that the 126-dimensional windowed tidal feature representation carries genuine predictive signal. The 47.9% MAE reduction of XGBoost relative to the median baseline (8,932 s vs. 17,168 s) establishes that the model is successfully learning the relationship between tidal phase history and slip timing, particularly for the dominant short-interval regime. This result validates the approach of using windowed tidal state features for this regression task and supports the use of XGBoost predictions as a practically useful tool for bulk ice stream mass flux estimation and tidal-cycle slip occurrence assessment [4], [11].

The regime-split breakdown reveals the structural limitation predicted in the introduction. XGBoost's MAE on long-interval events (11,089 s) is 42.7% higher than on short-interval events (7,769 s), and this gap is consistent across all three classical methods. This is not attributable to class imbalance alone: even after regime-aware routing, the long-interval regressor faces a sparser training set where the relationship between tidal phase and slip timing is weaker and more irregular, driven by the nonlinear interaction between tidal amplitude, residual stress state, and the slowly evolving mechanical state of the ice bed [7], [9]. The sequential binary split architecture of tree-based methods cannot efficiently represent these higher-order cross-feature interactions [17], [20], producing the observed performance asymmetry. Ridge regression's comparatively weaker overall performance (MAE: 9,702 s) confirms that a purely linear model is insufficient for this prediction task, consistent with the nonlinear tidal forcing structure.

4.3 Noiseless QRC Simulation

The noiseless FC-TFI quantum reservoir ensemble achieved an overall test MAE of 8,999 s, within 0.7% of the best classical model. This result should not be interpreted as a null finding. Theoretical work has established that noiseless quantum kernels span function classes comparable to classical kernel methods on structured tabular data [32], and parity in the noiseless setting is therefore the expected outcome rather than a surprising one. The relevant comparison is not noiseless QRC versus classical methods, but hardware QRC versus classical methods a comparison that requires the structured regularization provided by physical qubit decoherence.

Additionally, the low variance across the five ensemble iterations (test MAE range: $9059 - 9215\,\mathrm{s}$) suggests that the FC-TFI reservoir dynamics are stable under random Hamiltonian resampling, and that the noiseless performance ceiling is not primarily limited by reservoir expressiveness. This is consistent with the theoretical preference for the FC-TFI architecture: disordered inter-spin couplings that break permutation symmetry maximize memory capacity [26], while the Hamiltonian symmetry structure suppresses the exponential concentration of Pauli observables that would otherwise render the reservoir uninformative [27].

4.4 Toward Hardware Evaluation

The combined evidence from the classical regime-split analysis and the noiseless QRC validation performance motivates the hypothesis that hardware execution will produce a measurable advantage over the classical baseline, specifically on long-interval events where classical representational limitations are most acute. This prediction is grounded in prior work demonstrating that physical qubit decoherence on IBM Heron hardware produces better-than-noiseless performance on structured tabular prediction tasks [30], an effect absent in simulation and not seen with classical noise injection. Whether this advantage generalizes to tidally-forced geophysical data, with its specific combination of small catalog size, non-stationarity, and physically structured bimodal targets, is an important empirical question this work targets.

5 Conclusion

Whillans Ice Plain represents a unique, dynamic ice stream whose net displacement is delivered almost entirely through discrete tidally-paced stick-slip events. These events provide information about basal friction, subglacial water pressure, and proximity to shutdown for the ice stream. Accurate prediction of inter-event timing is both a practical tool for ice dynamics modeling and a window into physical processes that remain incompletely understood, including rupture nucleation phase, inter-event stress transfer, and the skipped-slip mechanism driving long-term deceleration. To approach this prediction problem, we first trained an XGBoost classifier to separate the 11-year [4] catalog into two physically motivated regimes, split at a threshold of 65,000 s corresponding to the trough of the bimodal inter-event interval distribution. This separation is physically grounded: the two regimes correspond to event clustering around 12-hour and 24-hour cycles, consistent with the known spring-neap tidal forcing structure at WIP [3], [9]. The classifier achieved a test accuracy of 83.9% (F1: 0.88 for the short regime, 0.77 for the long regime), with lower long-regime performance reflecting the physical ambiguity of neap-cycle events whose slip decision depends on subglacial conditions not directly observable in the tidal feature set.

With regime routing in place, three classical machine learning models were evaluated against MAE and RMSE on the held-out test set, each using a 126-dimensional windowed feature representation of 20 prior events plus the current event. All three models substantially outperformed both the median baseline (MAE: 17,168 s) and the persistence baseline (MAE: 22,803 s), confirming that the windowed tidal feature representation carries genuine predictive signal. XGBoost achieved the lowest error on both metrics (MAE: 8,932 s, RMSE: 16,888 s), a 47.9% reduction relative to the median baseline, and was therefore selected as the classical model in the hybrid quantum pipeline.

Since XGBoost gave the lowest classical error, it was used as the readout layer in the quantum-classical hybrid pipeline. The quantum reservoir was physically motivated in its design: input features were encoded via R_y rotation gates, data re-uploading was applied to access higher-order trigonometric feature interactions, and the reservoir dynamics were governed by a fully connected transverse-field Ising Hamiltonian approximated via first-order Trotterization. In noiseless simulation, the FC-TFI ensemble achieved a test MAE of 8,999 s and RMSE of 16,847 s, within 0.7% of the classical XGBoost result. This parity is the expected outcome: theoretical work has established that noiseless quantum kernels span function classes comparable to classical methods [32], and the advantage of hardware decoherence as a structured regularizer is absent in simulation.

The noisy simulation results gave an ensemble test MAE of 9561 s and RMSE of 17,516 s, consistent with prior findings that simulated noise gave no improvement over classical performance, but the literature suggests quantum advantage could be seen if run on quantum hardware. [38].

The results of this study make three contributions to the literature. First, they establish the first machine learning benchmark, classical or quantum, for time-to-next-slip regression on the [4] 11-year WIP catalog, providing a reproducible baseline for future work in cryoseismic prediction. Second, they demonstrate that the structural failure of classical tree-based methods on long-interval neap-regime events is empirically confirmed and physically interpretable, motivating richer nonlinear feature representations for this class of geophysical regression problem. Third, they introduce a physically motivated quantum-classical pipeline whose architecture is grounded in the documented bimodal structure of the WIP system, and contribute an empirical data point to the open question of where QRC hardware advantage materializes on real scientific data.

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